

CES Aggregator: Productivity and Price Effects Analysis

Setup: CES Production Function and Cost Minimization

Production Function:

$$Y = A \left[\sum_{i=1}^n \alpha_i X_i^\rho \right]^{\frac{1}{\rho}}$$

Cost Minimization Problem:

$$\min_{X_1, \dots, X_n} \sum_{i=1}^n p_i X_i \quad \text{subject to} \quad Y = \text{constant}$$

Optimal Input Demands:

$$X_i = \left(\frac{\alpha_i}{p_i} \right)^\sigma \left(\sum_{k=1}^n \alpha_k^\sigma p_k^{1-\sigma} \right)^{-1} Y$$

Where $\sigma = \frac{1}{1-\rho}$ is the elasticity of substitution.

Question 3: What happens to demand for input j when X_k increases?

Scenario: Exogenous increase in input k quantity

This represents a situation where the quantity of input k is increased exogenously (perhaps due to policy, availability constraints, or bundling), and we analyze how this affects the optimal demand for input j.

Approach: Conditional Input Demand

We need to find the conditional demand for input j given that X_k is fixed at some level.

The modified cost minimization problem becomes: $\min_{X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_n} \sum_{i \neq k} p_i X_i + p_k \bar{X}_k$

$$\text{Subject to: } Y = A \left[\alpha_k \bar{X}_k^\rho + \sum_{i \neq k} \alpha_i X_i^\rho \right]^{\frac{1}{\rho}}$$

Where $\bar{X}_k$ is the exogenously fixed level of input k.

Solving for conditional demand of input j:

The first-order condition for input j gives: $p_j = \lambda \cdot A \cdot \frac{1}{\rho} \left[\alpha_k \bar{X}_k^\rho + \sum_{i \neq k} \alpha_i X_i^\rho \right]^{\frac{1}{\rho}-1} \cdot \alpha_j X_j^{\rho-1}$

Where λ is the Lagrange multiplier (marginal cost).

After algebraic manipulation, the conditional demand for input j is:

$$X_j = \left(\frac{\alpha_j}{p_j} \right)^\sigma \left(\frac{Y}{A} \right)^{\sigma-1} \left[\alpha_k \bar{X}_k^\rho + \sum_{i \neq k} \alpha_i^\sigma p_i^{1-\sigma} \left(\frac{Y}{A} \right)^{(\sigma-1)\rho} \right]^{\frac{1-\sigma}{\rho}}$$

Effect of increasing X_k on demand for j:

Taking the derivative: $\frac{\partial X_j}{\partial X_k} = \left(\frac{\alpha_j}{p_j} \right)^\sigma \left(\frac{Y}{A} \right)^{\sigma-1} \cdot \frac{1-\sigma}{\rho} \cdot [\dots]^{\frac{1-\sigma}{\rho}-1} \cdot \alpha_k \rho \bar{X}_k^{\rho-1}$

Simplifying the sign analysis: $\text{sign} \left(\frac{\partial X_j}{\partial X_k} \right) = \text{sign} \left(\frac{1-\sigma}{\rho} \cdot \rho \right) = \text{sign}(1 - \sigma)$

Results:

Case 1: $\sigma > 1$ (High Substitution, $\rho > 0$)

- $1 - \sigma < 0$, so $\frac{\partial X_j}{\partial X_k} < 0$
- When X_k increases, demand for j DECREASES
- Strong substitution: more of k means less need for j

Case 2: $\sigma < 1$ (Low Substitution/Complements, $\rho < 0$)

- $1 - \sigma > 0$, so $\frac{\partial X_j}{\partial X_k} > 0$
- When X_k increases, demand for j INCREASES
- Complementarity: more of k increases the marginal productivity of j

Case 3: $\sigma = 1$ (Cobb-Douglas, $\rho = 0$)

- $\frac{\partial X_j}{\partial X_k} = 0$
- Demand for j is independent of X_k quantity
- No interaction between input quantities

Question 1: Productivity Increase of Input k

Scenario: Productivity of input k increases

This can be modeled as an increase in α_k (the distribution parameter for input k).

Effect on Demand for Input j ($j \neq k$):

From the optimal demand equation:

$$X_j = \left(\frac{\alpha_j}{p_j} \right)^\sigma \left(\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma} \right)^{-1} Y$$

Taking the derivative with respect to α_k :

$$\begin{aligned} \frac{\partial X_j}{\partial \alpha_k} &= \left(\frac{\alpha_j}{p_j} \right)^\sigma Y \cdot \frac{\partial}{\partial \alpha_k} \left(\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma} \right)^{-1} \\ &= - \left(\frac{\alpha_j}{p_j} \right)^\sigma Y \cdot \frac{\sigma \alpha_k^{\sigma-1} p_k^{1-\sigma}}{\left(\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma} \right)^2} \end{aligned}$$

Since all terms are positive, we have:

$$\frac{\partial X_j}{\partial \alpha_k} < 0$$

Result 1: When productivity of input k increases, demand for input j DECREASES

Economic Intuition by Substitution Elasticity:

- **High substitution** ($\sigma > 1, \rho > 0$) : Strong substitution effect - firms substitute away from other inputs toward the more productive input k
- **Low substitution** ($\sigma < 1, \rho < 0$) : Weak substitution effect, but scale effect dominates - overall cost reduction leads to expansion, but input k still gains relatively more

Question 2: Price of Input k Falls

Effect on Cost Share of Input j ($j \neq k$):

The cost share of input j is: $s_j = \frac{p_j X_j}{\sum_{i=1}^n p_i X_i} = \frac{\alpha_j^\sigma p_j^{1-\sigma}}{\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma}}$

Taking the derivative with respect to p_k : $\frac{\partial s_j}{\partial p_k} = - \frac{\alpha_j^\sigma p_j^{1-\sigma} \cdot \alpha_k^\sigma (1-\sigma) p_k^{-\sigma}}{\left(\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma} \right)^2}$

Simplifying: $\frac{\partial s_j}{\partial p_k} = - \frac{(\sigma-1) \alpha_j^\sigma p_j^{1-\sigma} \alpha_k^\sigma p_k^{-\sigma}}{\left(\sum_{i=1}^n \alpha_i^\sigma p_i^{1-\sigma} \right)^2}$

The sign depends on $(\sigma - 1)$:

Case 1: $\sigma > 1$ (High Substitution, $\rho > 0$)

- $\sigma - 1 > 0$, so $\frac{\partial s_j}{\partial p_k} < 0$
- **When p_k falls, s_j INCREASES**
- Despite substitution toward k, j's cost share rises due to scale effects

Case 2: $\sigma < 1$ (Low Substitution, $\rho < 0$)

- $\sigma - 1 < 0$, so $\frac{\partial s_j}{\partial p_k} > 0$
- **When p_k falls, s_j DECREASES**
- Limited substitution means j loses cost share as k becomes relatively cheaper

Case 3: $\sigma = 1$ (Cobb-Douglas, $\rho = 0$)

- $\frac{\partial s_j}{\partial p_k} = 0$
- **Cost share of j remains constant regardless of price changes to k**

Summary of Results

Scenario	Effect	Condition
Productivity of k increases	Demand for j decreases	Always (substitution dominates)
Price of k falls	Cost share of j increases	$\sigma > 1$ (high substitution)
Price of k falls	Cost share of j decreases	$\sigma < 1$ (low substitution)
Price of k falls	Cost share of j unchanged	$\sigma = 1$ (Cobb-Douglas)
Quantity X_k increases	Demand for j decreases	$\sigma > 1$ (high substitution)
Quantity X_k increases	Demand for j increases	$\sigma < 1$ (low substitution/complements)
Quantity X_k increases	Demand for j unchanged	$\sigma = 1$ (Cobb-Douglas)

Economic Intuition

Productivity Effect:

Higher productivity of input k makes it relatively more attractive, leading firms to substitute away from other inputs toward k.

Quantity Effect on Demand:

When the quantity of input k increases exogenously:

- **High substitution ($\sigma > 1$):** Demand for j decreases because inputs are substitutes - having more k reduces the need for j

- **Low substitution ($\sigma < 1$):** Demand for j increases because inputs are complements - having more k makes j more productive and valuable
- **Unit substitution ($\sigma = 1$):** Demand for j is unaffected - inputs are independent in the Cobb-Douglas case

This reveals the fundamental complementary vs. substitutional relationships embedded in the CES structure.